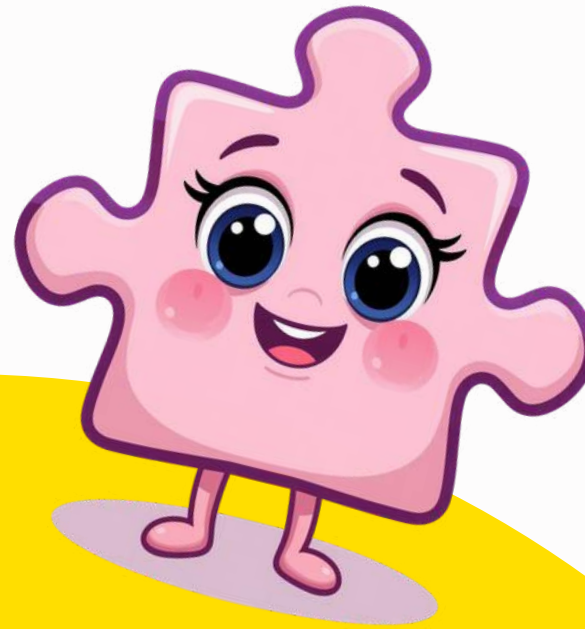


Maths & Science Olympiad Challenge



Guided Sprint: Olympiad Hack Challenge

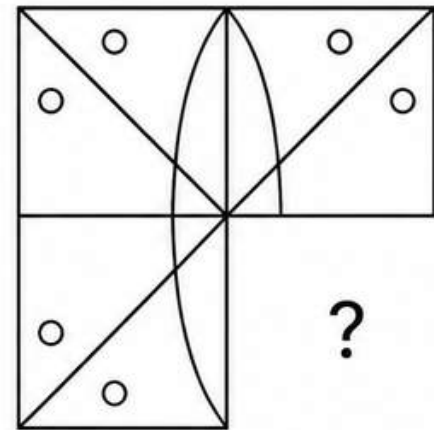
Can you solve this multi-step puzzle?
Combine patterns, particles, and logic!
Strategy tip: Use elimination and look
for clues carefully in each step.



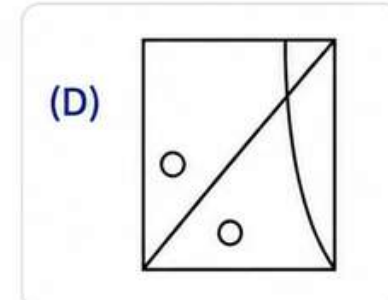
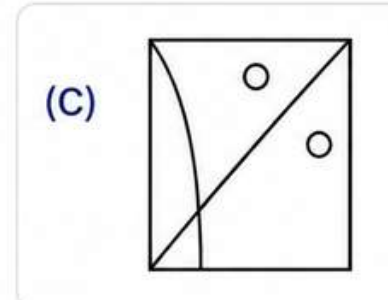
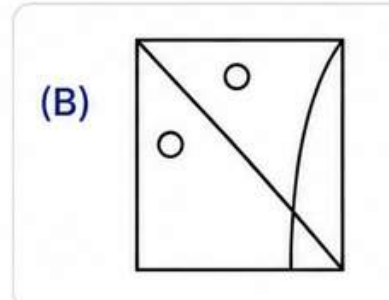
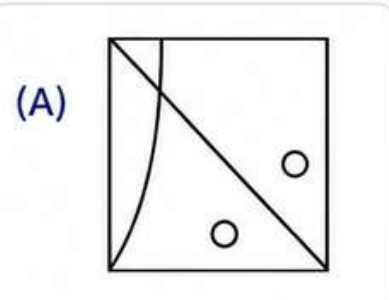
Decoding Patterns: Rangoli & Rotations

Example 1

Find the missing figure that completes the pattern.



OPTIONS



Shape Changes

Watch how shapes transform! Triangles become squares, circles grow bigger.



Rotation Rules

Triangle rotates 90° clockwise each step. Track the direction!



Shading & Dots

Dots increase by 2 each time. Count carefully to predict the next!



Pattern spotted! Rotate. Count. Predict.

Arithmetic Patterns:

The Indian Coins Puzzle



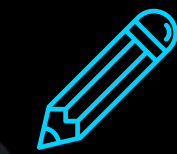
Different Ways

How many combinations can make ₹50?
Let's explore all the possibilities!



Counting Totals

Add up your coins: $10 \times ₹5 = ₹50$, or mix
₹5 and ₹2 coins together!



Systematic Approach

Start with max ₹5 coins, then swap for ₹2
coins step by step. Pattern spotted!





Solid & Liquid

Molecules: Tawa & Bowl

Solids hold shape, liquids flow and fill containers. Iron Tawa molecules are tightly packed and vibrate in place. Steel Bowl with water has loosely packed molecules that flow freely!

Gas Molecules: Agarbatti Aroma Magic

Gas particles move fast and fill the air!
When you light an agarbatti, the smoke
spreads everywhere in the room.
No shape, no fixed volume!
Particles zoom freely in all directions.





Logical Reasoning

Five Friends at the Bus Stop

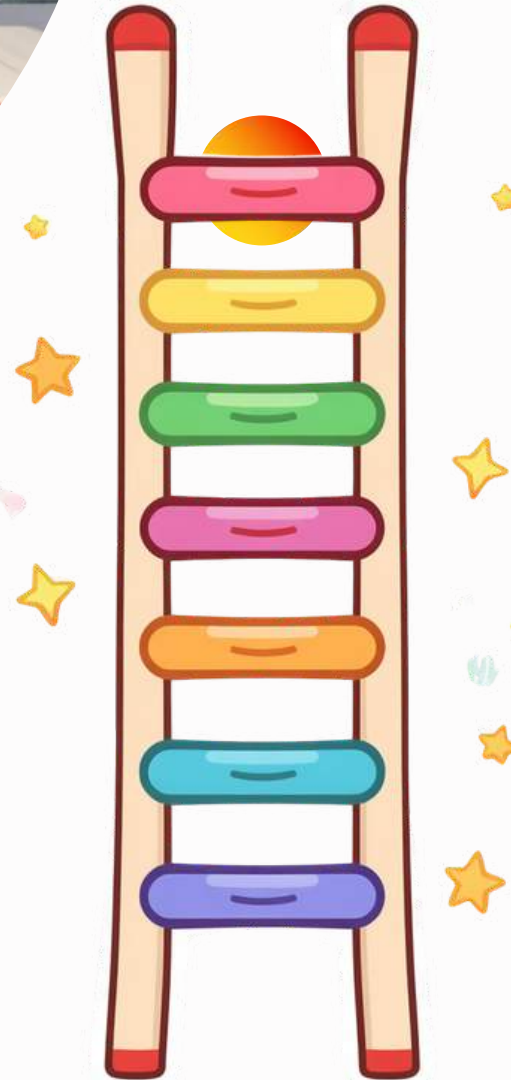
FIND OUT WHO'S THE TALLEST!

Five students-Arjun, Bhavana, Charan, Diya, and Esha-are waiting sequentially in a line for a school bus at a designated stop in Bengaluru. The following structural parameters are given:

1. Arjun is positioned strictly taller than Charan, but structurally shorter than Diya.
2. Bhavana is verified to be shorter than Charan.
3. Diya is explicitly documented to be shorter than Esha.

Shep Seating Matrix

Arjun	Bhavan	Esiay
Arjun	Bharan	Diyan
Arijuna	Elavan	Esha



Logical Reasoning Strategy

- Step 1: Read all clues carefully first
- Step 2: Start with the most definite clue
- Step 3: Eliminate impossible options one by one
- Step 4: Check your answer against all clues

Use a simple chart or ladder to track what you know. Cross out wrong answers as you go. Excellent deduction! Keep going!

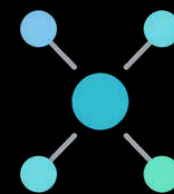


Session Wrap-Up & Mission



Pattern Analysis

Spot sequences, predict what comes next, and decode visual patterns like a pro!



Particle Properties

Understand how solids, liquids, and gases behave at the molecular level.



Logical Elimination

Use clues and systematic thinking to solve tricky puzzles step by step.



Thank You!

Let's Crack the Code!

Ready for your next Olympiad adventure?
You are now pattern detectives and science explorers!

